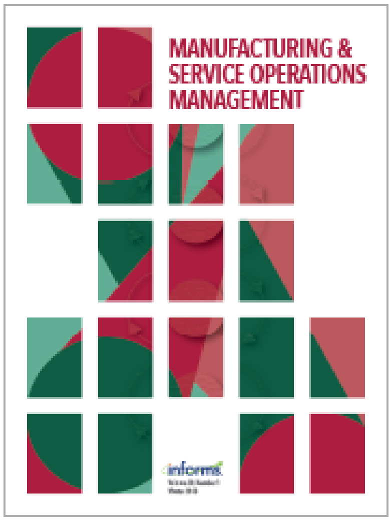

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Estimating the Stockout-Based Demand Spillover Effect in a Fashion Retail Setting

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Abstract. ***Problem definition***: In brick-and-mortar fashion retail stores, inventory stockouts are frequent. When a specific size of a fashion product is out of stock, the unmet demand might not be completely lost because of spillovers to adjacent sizes of the same style or to other styles. Little research has been done to study consumer response to stockouts of fashion products because researchers had limited access to proprietary data of fashion retailers and because it is challenging to estimate stockout-based demand spillover patterns using existing approaches due to the enormous number of stockkeeping units (SKUs) and frequent stockouts in fashion retail stores. To fill this void in the literature, we empirically estimate the stockout-based demand spillover effect in a fashion retail setting. ***Methodology/results***: We obtain a large-scale data set from a fashion retail chain selling world-renowned sportswear brands. The retail stores in the sample are dedicated to products of a single brand. Using around 1.5 million granular and real-time sales and inventory records of 217 stores, 503 men's footwear products, and 4,024 SKUs over a two-year period, we develop a difference-in-differences framework to estimate the stockout-based cross-size demand spillover effect. We demonstrate the validity of this framework by conducting a pretrend test and a placebo test. We find that roughly 51.7% of the unmet demand of an out-of-stock SKU spills over to adjacent sizes of the same style when they are in stock: 25.1% to the adjacent-larger size and 26.6% to the adjacent-smaller size. The cross-size demand spillover effect is larger in regular stores than in flagship stores, larger for casual sports shoes than for specialized sports shoes, and larger for low-price products than for high-price products. Adapting an existing attribute-based demand model to our setting, we estimate that roughly 20.2% of the unmet demand of an out-of-stock SKU spills over to different styles when they are in stock. Taken together, these estimations suggest that about 28.1% of the unmet demand of an out-of-stock SKU becomes lost sales. We further find that when stockouts are widespread among SKUs, stockout-based demand spillovers are significantly reduced, resulting in much increased lost sales. ***Managerial implications***: First, we empirically quantify the stockout-based cross-size demand spillover effect and its impact on lost sales in a brick-and-mortar fashion retail setting. Second, our simulation analysis shows that incorporating the cross-size demand spillover effect into the sportswear retail chain's proactive transshipment decision can substantially reduce its transshipment cost and improve its profitability.

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Keywords: retail operations • inventory stockouts • demand substitution • lost sales • proactive transshipment

1. Introduction

Stockouts are widespread in the retailing world. By combining results from 40 prior studies on retail stockouts, Gruen et. al. (2002) find that the worldwide average stockout rate was 8.3% for 18 consumer good product categories ranging from salty snacks to diapers to hair care. The stockout rate varies significantly across product categories and regions (Gruen and Corsten 2008, Sanchez-Ruiz et al. 2018). There are four typical consumer responses to stockouts: stockout-based substitutions within or between brands at the same store, delayed purchase, no purchase, or purchase at another store. One U.S. consumer survey study shows that for 11 consumer packaged goods, on average 40% of the 23,000 surveyed consumers will substitute within or between brands at the same store when facing stockouts (Gruen et al. 2002). Researchers in operations management, marketing, and economics have developed models to estimate the effects of stockouts on consumer demand (Anupindi et al. 1998, Musalem et al. 2010, Conlon and Mortimer 2013). The general finding is that stockout-based substitutions are significant and ignoring them may lead to biased estimates of demand.

The aforementioned stockout studies almost exclusively focus on grocery retail settings, where point-of-sale scanner data have become widely available to academic researchers. However, few studies have systematically documented stockout patterns in fashion retail settings and empirically studied consumer response to stockouts of fashion products. After surveying the literature, we find only one study that systematically explores inventory stockouts in a fashion retail setting. Using sales and inventory data of men's T-shirts with 32 stockkeeping units (SKUs) in a department store, Kalyanam et al. (2007) report that the average percentage of out-of-stock weeks in a quarter for an SKU was 3.37% but varied widely from 0% to 76.92%. They find that depending on which item is out of stock, the effect of stockouts on the sales of the entire product category may be positive or negative. The paucity of research on fashion products is partly due to academic researchers' inaccessibility to proprietary data of fashion retailers.[1] Some anecdotal evidence suggests that stockout rates can be much higher in fashion retail settings than in other retail settings because fashion products typically have short life cycle, high demand uncertainty, and long lead time that prevents in-season replenishment, thereby leading to high costs of stockouts because of lost sales (Fisher et al. 1994).

One feature that distinguishes fashion products from other consumer goods is the size dimension. Most fashion products, for example, apparel and footwear, come in a range of different sizes. For a consumer, when a product of a particular style is available, but not in his size, he faces a stockout situation. When a specific size of a fashion product is out of stock, the unmet demand might not be completely lost because of consumer substitution to adjacent sizes of the same product or substitution to products with different styles (designs or colors).

In fashion retail settings, the meaning of demand substitution can be subtle. Unlike in grocery retail settings, a consumer might not realize that he purchases a substitute product because he does not necessarily have an ideal product in mind before entering a fashion retail store but ends up buying his most preferred product available there. However, from the retailer's perspective, regardless of whether a consumer has an ideal product in mind ex ante, he can figure out his best size-style SKU after observing the stores' displayed product assortment and trying on some items. For example, he can figure out that size 9 is his best size by trying on a red-colored shoe in size 9 even though his most preferred choice is a blue-colored one in size 9, which is out of stock. In this case, the consumer might buy his second-best size-style SKU (e.g., size 9.5 in blue color), which will experience a sales increase as a result. To be precise, we use the term "stockout-based demand spillover" instead of "demand substitution" for the rest of the paper to refer to this sales increase effect induced by inventory stockouts in fashion retail settings.

Consumer response to inventory stockouts can be substantially different in fashion retail settings than in grocery retail settings, and it is not ex ante clear whether consumers are more or less willing to purchase substitute products. On the one hand, consumers' visits to fashion retail stores are much less frequent than their visits to grocery stores, which could make them more willing to purchase substitutes instead of walking out of stores empty-handed. On the other hand, fashion products are generally discretionary and more expensive than food or household items sold in grocery stores, which could make consumers less willing to settle for something other than their most preferred. Most importantly, unlike food or household items, fashion products will be worn by consumers and carry a long-lasting effect on their personal image. Therefore, consumers might be reluctant to settle for something else when their most favorable sizes or styles are out of stock. The scarcity of relevant studies in the academic literature prompted us to empirically assess the stockout-based demand spillover effect in a fashion retail setting.

To this end, we obtained a large-scale data set from one of the largest sportswear fashion retail chains in China. This retailer sells products of several world-renowned sportswear brands. Most of the retailer's stores are brand-dedicated stores, that is, each store carries a single brand. One of the main product categories of this retailer is men's sports footwear. We choose to focus on this product category for several reasons. First, stockouts are frequent in this product category. The average stockout rate is 15.9% at the SKU-store-day level over the product life cycle. For SKUs with less common men's shoe sizes (e.g., size 6.5 or 10), the average stockout rate can exceed 20%. Second, footwear is one of the largest categories of fashion products, totaling sales of $366 billion worldwide in 2019 (Lissaman 2019). Third, academic studies on footwear products are, to the best of our knowledge, almost nonexistent. Last but not least, unlike apparel whose inventory is usually completely or partially displayed on the shop floor and visible to consumers, footwear products have a store layout that only displays samples on the shop floor and holds inventory in the back room. As a result, consumers need to check with sales associates to get access to the SKUs they wish to try on and then decide whether to purchase. Hence, this footwear retail setting allows us to eliminate the "broken assortment" effect (Caro and Gallien 2010) and the "assortment attractiveness" effect (Kalyanam et al. 2007), which exist in retail settings where inventory is displayed on the shop floor and visible to consumers. Without the interference of these two effects, we can estimate the true stockout-based demand spillover effect.

Specifically, we develop a difference-in-differences (DID) framework to estimate the stockout-based cross-size demand spillover effect using granular and real-time sales and inventory records of 217 stores, 503 men's footwear products, and 4,024 SKUs over a two-year period. The granularity of the data allows us to distinguish in-stock days from out-of-stock days for individual SKUs (a unique size-style combination) in each store. We view a stockout event of an SKU as a treatment and identify it at the SKU-store-day level. Accordingly, we define the treatment group as the set of SKUs affected by the stockout event and define the control group as the set of SKUs unaffected by the stockout event. We demonstrate the validity of this framework by conducting a pretrend test and a placebo test.

We find that roughly 51.7% of the unmet demand of an out-of-stock SKU spills over to adjacent sizes of the same style when they are in stock: 25.1% to the adjacent-larger size and 26.6% to the adjacent-smaller size. The cross-size demand spillover effect is larger in regular stores than in flagship stores, larger for casual sports shoes than for specialized sports shoes, and larger for low-price products than for high-price products. Adapting the attribute-based demand model by Fisher and Vaidyanathan (2014) to our setting, we estimate that roughly 20.2% of the unmet demand of an out-of-stock SKU spills over to different styles when they are in stock. Taken together, these estimations suggest that about 28.1% of the unmet demand becomes lost sales. We further find that when stockouts are widespread among SKUs, stockout-based demand spillovers are significantly reduced, resulting in much increased lost sales.

Based on our empirical estimations, we conduct a simulation analysis on the sportswear retail chain's interstore proactive transshipment decision problem. Specifically, we build a mathematical programming model for a single-period, multisize product's proactive transshipment problem and use field data to conduct a simulation analysis. We find that incorporating the cross-size demand spillover effect into the proactive transshipment decision problem can reduce our retail partner's transshipment cost by 7.1%–24.9% and improve its profit by 0.1%–1.0%.

Although there is a sizeable literature on the effects of stockouts on retail sales in the field of operations management, marketing, and economics, as far as we know, the existing literature has seldom considered fashion products. Studying the stockout effect of fashion products faces the challenge of the enormous number of SKUs and frequent stockouts in brick-and-mortar retail stores. The typical empirical setting in the existing literature has less than 50 SKUs (Musalem et al. 2010). However, a fashion retailer can sell hundreds or even thousands of SKUs for just its footwear category at the store level. Therefore, the methodologies developed in the existing literature cannot be easily applied to a fashion retail setting. Our work tries to fill this theory-practice gap and makes two main contributions:

First, we take advantage of our retail partner's large-scale data set and develop a DID framework to estimate the stockout-based cross-size demand spillover effect. The retailer's granular and real-time sales and inventory records enable us to track inventory level and stockouts at the SKU-store-day level, thereby allowing us to assess the more granular demand spillover effect of stockouts than the existing literature (e.g., Kalyanam et al. (2007) study the quarterly stockout effect). A notable advantage of our method is the fact that it does not need to assume a stable set of consumer choice alternatives or impose assumptions on the total market demand size (Musalem et al. 2010) or market share (Vulcano et al. 2012), which are often required when applying choice-model based estimation approaches. Moreover, many attributes of fashion products are not digitized—the complex colors and designs of fashion products are often recorded in a textual and incomplete format in a retailer's database, making attribute-based estimation methods not easily adaptable to fashion products. Our estimation framework avoids this challenge because it does not use product attributes.

Second, our work demonstrates a strong cross-size demand spillover effect in a brick-and-mortar fashion retail setting. Our result indicates that a significant portion of consumers are willing to settle for an adjacent size when their most preferred size is unavailable. To the best of our knowledge, our paper is the first to empirically quantify consumer response to stockouts at the SKU level in a fashion retail setting. Our simulation analysis further shows that incorporating the cross-size demand spillover effect can enable our retail partner to improve its interstore proactive transshipment, a tactic many traditional fashion retailers rely on to match supply with demand during the short selling season.

The rest of this paper is organized as follows. Section 2 reviews the related literature, and Section 3 describes the empirical setting. We present our methodology and estimation results in Sections 4 to 6 and a simulation analysis in Section 7. We conclude in Section 8.

2. Literature Review

Our work contributes to the literature that studies the effects of stockouts on consumer demand. This literature has gained an increasing attention from researchers in operations management, marketing, and economics in recent years. We categorize studies in this literature into three streams according to their methodologies: (1) estimations based on a product substitution model; (2) estimations based on a multinomial logit (MNL) choice model; and (3) experiments. Table 1 provides a list of representative papers in this literature. We will next elaborate on each of the three streams and review related papers.

The first stream of the stockout literature adopts product substitution models to estimate the effects of stockouts. Studies in this stream assume an exogeneous stockout-based substitution pattern between different products and estimate substitution probabilities using sales and inventory availability data. One of the earliest studies in this stream is Anupindi et al. (1998), who assume a Poisson demand process and use an exogenous product substitution matrix to estimate demand spillovers caused by stockouts. They apply their estimation method to daily sales and inventory data collected from vending machines and estimate the demand spillover rates of six soft drink SKUs. Similarly, Kalyanam et al. (2007) use a product substitution model to estimate the cross-item stockout effects of 32 men's T-shirt SKUs sold in a department store. What differentiates the two studies is the fact that Kalyanam et al. (2007) estimate the effect of an item's stockout on the entire category sales by constructing an *assortment attractiveness* index, which can be positively or negatively affected by an item's stockout. However, the method of Kalyanam et al. uses an aggregate stockout measure that is the percentage of weeks that a T-shirt SKU is out-of-stock in a store in a quarter. In other words, it uses a quarterly stockout measure. The lack of granular stockout information in this study makes it impossible to assess the SKU-level stockout effect in retail stores, which we estimate in our study.

Table 1. Summary of the Literature on the Effects of Stockouts

Studies	Empirical context	No. of SKUs	Stockout measure	Time units	Sample period	Methodology
Anupindi et al. (1998)	Vending machines: soft drinks	6	SKU-level inventory status	Daily	9 months	Product substitution model
Kalyanam et al. (2007)	Department stores: men's T-shirts	32	No. of stockout weeks in a quarter	Quarterly	2 years	Product substitution model
Campo et al. (2003)	Grocery stores: cereals, margarines	24	Estimated	Weekly	28 weeks	MNL choice model
Musalem et al. (2010)	Grocery stores: shampoos	24	SKU-level inventory status	Daily	15 days	MNL choice model
Jing and Lewis (2011)	Online grocery store	–	Product category	Weekly	14 months	MNL choice model
Vulcano et al. (2012)	Airline/retail sales	15/6	N/A	Daily/weekly	7 days/8 weeks	MNL choice model
Che et al. (2012)	Single grocery store	6	Estimated	Consumer shopping trip	2 years	MNL choice model
Conlon and Mortimer (2013)	Vending machines: snacks	44	SKU-level inventory status	Every 4 hours	1 year	MNL choice model
Fitzsimons (2000)	Laboratory experiment	6	N/A	N/A	N/A	Experiment
Anderson et al. (2006)	Mail-order catalog retailer: bedding and home accessaries	5	Item fullfillment status in an order	N/A	5-week treatment	Experiment
Huang and Zhang (2016)	Experiment on Amazon Mechanical Turk: laundry detergents	3	N/A	N/A	N/A	Experiment

Despite being a highly intuitive and interpretable model, the product substitution estimation approach faces a major computational challenge when applied to a fashion retail setting because of the curse of dimensionality. This is because for N number of SKUs, N^2 spillover rates must be estimated, and there are 2^N possible states representing the various possible stockout conditions. Kalyanam et al. (2007) study 32 men's T-shirt SKUs in a department store, with a total of eight colors and four sizes. This product category, because of its relatively simple style and small size assortment, probably has the lowest number of SKUs among all fashion products. A typical fashion product category, however, may have hundreds or even thousands of SKUs with different designs, colors, and sizes. In our empirical setting of men's sports footwear products, a typical store carries over several thousand SKUs in a year.

To overcome the curse of dimensionality, the second stream of the stockout literature adopts the multinomial logit (MNL) choice model or some variations of it to estimate the stockout effect. These models use product attributes to construct consumer utility functions. One challenge of applying this approach to a fashion retail setting is the lack of accurate and complete attribute data for fashion products in retailers' databases because it is hard to translate the textual and incomplete description of a fashion product into digital data. For example, the design and color of a cocktail dress or a sports shoe cannot be easily described with just a few words, let alone numbers. Moreover, the MNL choice model assumes that the effects of product features on consumer utility are additive. In reality, many attributes are highly correlated. For example, the matching between a color and a design cannot be simply captured by the sum of the effect of a color and the effect of a design. Therefore, fashion products' attributes are inherently complex and high dimensional. To this end, machine learning–based feature extraction methods using product images are under development and yet to find their uses both in academic research and in practice.

Another challenge of using the MNL choice model estimation approach is to track inventory status for all customer purchases. Consumer choice sets are changed by stockout events and need to be inferred from inventory data. However, this information is usually not readily available from retailers' existing databases. Therefore, researchers develop various methods to get around the issue. For example, Conlon and Mortimer (2013) use a random coefficient MNL model to estimate the effects of stockouts and develop an expectation-maximization algorithm for model estimation. Musalem et al. (2010) adopt a similar model but use a different estimation approach in which the actual choice set of each consumer is generated by simulation. Vulcano et al. (2012) study the demand estimation problem in a multiproduct, multiperiod retail setting, in which each product's inventory status is known but the number of consumer arrivals is unavailable. They propose an efficient estimation procedure via likelihood function decomposition. Other researchers estimate a modified MNL choice model by incorporating an out-of-stock indicator to remove the out-of-stock item from the choice set (Campo et al. 2003, Che et al. 2012). Jing and Lewis (2011) analyze the impact of stockouts on consumer purchases in an online retail setting and find that stockouts might increase or decrease consumer purchase frequency depending on the product category.

The aforementioned estimation approaches are based on the MNL choice model, which can nicely capture consumer purchasing behaviors for a given assortment in a retail context. Despite the powerful capabilities and flexibility of the MNL choice model, it has two important restrictions. (1) It usually needs to impose strong assumptions on consumer choices, for example, independence of irrelevant alternatives (IIA). (2) Empirical estimations can be very challenging when a large number of SKUs are involved. Boada-Collado and Martínez-de-Albéniz (2020) show that in their setting with around 250 products/SKUs, the MNL estimation is already beyond the capability of the current R package. In a typical fashion retail setting, a store can carry hundreds or thousands of SKUs, which makes it infeasible to use the existing MNL-based estimation approaches. Recently, Jiang et al. (2020) develop a choice model to tackle the challenge of high-dimensional substitution pattern and relax the IIA assumption in an online retailing context. They introduce a flexible variance-covariance matrix for utility error terms in a probit model and propose a two-stage methodology for estimation. This new approach relies on the denser information in online customer clickstream data, which is not available in brick-and-mortar fashion retail settings.

The third stream of the stockout literature explores consumers' perception about stockouts via experiments. Fitzsimons (2000) finds that higher consumer commitment to a stockout item is associated with more negative response to stockouts in the form of lower satisfaction with the decision process and higher likelihood of switching stores on subsequent shopping trips. Anderson et al. (2006) conduct a large-scale field experiment with a catalog retailer. They find that stockouts reduce consumer purchase amount both in the short run and in the long run. Huang and Zhang (2016) conduct experiments through Amazon Mechanical Turk to study how consumers' product selection decisions are affected by stockouts and find that stockouts will increase the popularity of the products with similar attributes as the out-of-stock one.

Besides the aforementioned literature of stockouts, it is worth noting that Boada-Collado and Martínez-de-Albéniz (2020) show that sales of fashion products are positively affected by store inventory through better displays and increased category attractiveness. This positive effect of inventory is not due to the reduction of lost sales because the retailer has extremely low stockout rate (0.1%) for the studied two product categories: T-shirts and dresses.

3. Empirical Setting and Data

3.1. Company Background

Our retail partner is one of the largest sportswear retail chains in China. This company sells products of more than 20 different sportswear brands. It has grown rapidly and currently operates more than 1,000 retail stores in large cities throughout China. For the company's core business, it has a distributor relationship with two of the world's most famous sportswear brands, and the retail stores carrying these two brands are dedicated to selling a single brand. These brand-dedicated stores are where our data were collected from.

Our data set covers a two-year sample period from August 1, 2014, to July 31, 2016, during which the company operated 735 retail stores. We focus on 217 stores that sold products exclusively for brand A.[2] These stores had been operating for at least six months by the start of the sample period, and they sold 7,715 footwear products and 7,534 apparel products during the sample period. Footwear is the most important product category, contributing to around 60% of the retailer's total revenue. Our study focuses on this footwear product category.

3.2. Inventory and Sales Data

A unique feature of our data set is the *granularity* of its inventory and sales records. Specifically, the data come in the form of operational records that cover four key steps in store operations that affect inventory status: (1) new product arrivals, (2) customer sale transactions, (3) inbound inventory transshipments, and (4) outbound inventory transshipments. Inventory status is observable at the SKU-store level and updated in real time. As a result, we can accurately track inventory status of each SKU in each store.

Figure 1 illustrates the hierarchy of the retailer's footwear products. At the top level, the retailer carries both men's and women's footwear products. For each of the two product families, there are three major product categories: basketball, running, and casual. Each of the product categories may consist of several hundred different products. Each style is characterized by a unique design and color combination, whereas an SKU is a product with a specific style and a specific size. For men's footwear products, there are eight standard sizes ranging from 6.5 to 10 (U.S. size system).[3]

Figure 1. (Color online) Sports Footwear Product Hierarchy

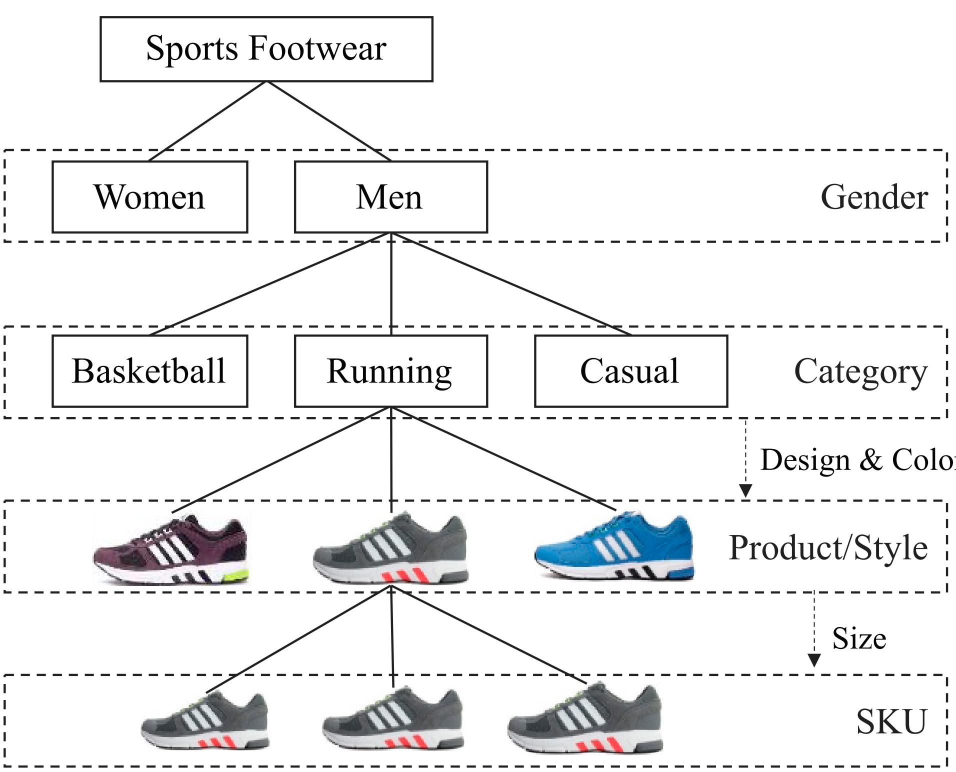


Table 2. Average Starting Inventory, Stockout Rate, and Daily Sales at the SKU-Store Level

	6.5	7	7.5	8	8.5	9	9.5	10
				Average starting inventory				
Basketball	1.0	1.4	2.0	3.9	5.1	4.6	5.3	3.4
Running	2.3	3.9	3.4	6.5	6.3	4.3	4.4	2.2
Casual	1.7	3.2	2.7	5.4	5.0	3.6	3.1	1.7
				Average stockout rate (%)				
Basketball	39.4	45.0	29.5	18.6	13.1	11.4	10.8	13.6
Running	31.3	19.5	18.3	12.8	10.2	11.7	13.8	25.4
Casual	36.4	22.3	20.6	12.1	10.8	11.2	15.6	24.8
				Average daily sales				
Basketball	0.008	0.071	0.053	0.064	0.070	0.058	0.065	0.044
Running	0.048	0.052	0.044	0.072	0.064	0.044	0.048	0.040
Casual	0.067	0.064	0.050	0.077	0.068	0.045	0.046	0.042

To provide an overview of the data set, we choose brand A's top 10% most sold men's footwear products, which consist of 55 basketball shoes, 173 running shoes, and 275 casual sports shoes. Table 2 displays the average starting inventory at the store-SKU level, and it is less than seven pairs. For those less popular sizes such as 6.5 and 10, the average starting inventory is around three pairs. This shallow inventory depth is reasonable given the limited back-room capacity, huge number of SKUs, and relatively short product life cycles. We define product life cycle as the number of days during which a store sells a product. The median life cycles of basketball, running, and casual shoes are around 75, 88, and 74 days, respectively. The length of these life cycles is slightly shorter than a physical season (90 days). This is expected because brand A introduces four major seasons of products annually.

Table 2 also displays the average stockout rate at the SKU-store level, which is defined as the number of stockout days of an SKU in a store divided by the respective product life cycle. Given the low starting inventory, the average stockout rate is fairly high. Take casual sports shoes as an example, the popular sizes, for example, 8, 8.5, and 9, have a stockout rate of around 11%. However, the stockout rate of less popular sizes, for example, 6.5, can be as high as greater than 35%. The high stockout rate is caused by the low starting inventory of these sizes in stores. Finally, Table 2 displays the average daily sales at the SKU-store level, which shows that sales events distribute sparsely across time. For a running shoe product in size 8, the average number of daily sales is only 0.072. Therefore, it takes approximately 14 days for a store to sell one unit of this SKU.

4. DID Estimation Framework for the Stockout-Based Cross-Size Demand Spillover Effect

In this section, we describe the conceptual framework of our DID approach for estimating the stockout-based cross-size demand spillover effect.

4.1. Store Layout of Footwear Products

Our retail partner's brand A store layout for its footwear products is illustrated in Figure A1 in Online Appendix A. Specifically, a single size for each style is displayed on the shop floor, whereas the inventory of different sizes is stored in the back room. Because of this layout, consumers can observe shoe samples on the shop floor and evaluate the esthetics of different styles but do not directly observe the inventory status of an SKU with a specific size. This type of store layout is common in footwear retail stores. It can avoid too many shoe boxes showing in the sales area and thus create spaciousness and neatness. It can increase the brand perception for retailers with a high-end positioning. Therefore, this layout is also often used in high-end fashion retail stores selling luxury apparels and accessories.

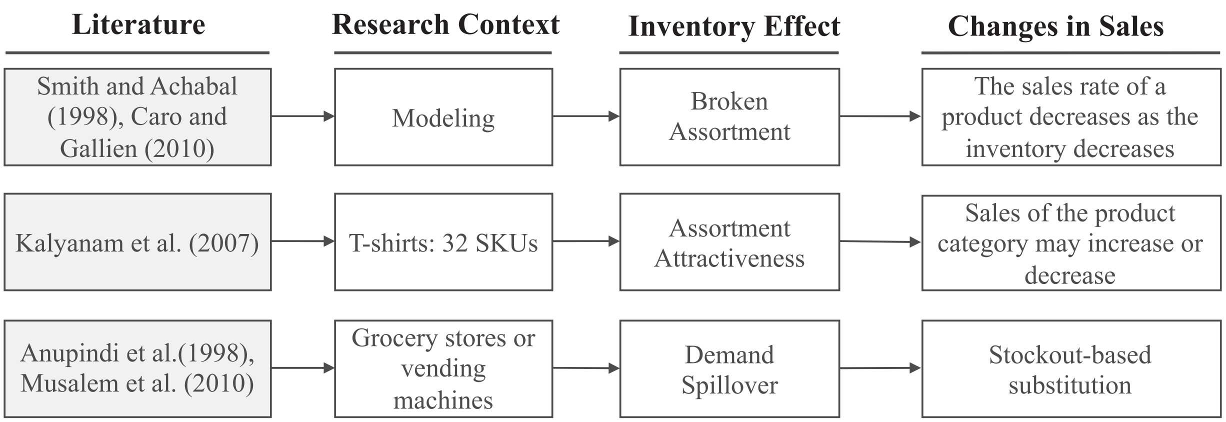
4.2. Effects of Inventory Stockouts on Sales

To develop our estimation framework, we draw on the findings from the literature that studies the effects of stockouts on sales. For the SKUs experiencing stockouts, obviously their own sales will be reduced. What is not clear is how the stockout of one product might affect the sales of other products. To answer this question, the stockout literature has identified three effects, as summarized in Figure 2.

First, Smith and Achabal (1998) find that the sales rates of apparel products may decrease when the available inventory does not cover all sizes or colors, which is the so-called "broken assortment" effect. This study uses apparel sales data from a department store to demonstrate a positive correlation between sales rates and inventory levels. According to Caro and Gallien (2010), Zara has the policy that retail stores should pull back the entire assortment of an apparel article from display when its selected key sizes are out of stock because the broken assortment effect will negatively affect the sales of the apparel article and the freed-up store space can be better used to sell other apparel articles. In our footwear retail setting, because inventory is held in the back room, consumers do not directly observe the inventory status of a product. In addition, as long as some sizes of the same product are available, one of them can be displayed on the shop floor. Therefore, there is no broken assortment effect in our setting.

Second, Kalyanam et al. (2007) discover that depending on which item in a product category is out of stock, the item's stockout may affect the whole product category's sales positively or negatively, which the authors call the "assortment attractiveness" effect. This study uses *quarterly* sales data of men's T-shirts from a U.S. retailer that has 196 stores. A total of 32 SKUs (color and size combinations) are considered. The authors find that the stockouts of most SKUs in the category

Figure 2. Effects of Inventory Stockouts on Sales



negatively affect the category sale because it reduces the attractiveness of the category. However, the stockouts of some SKUs positively affect the category sale because they increase the attractiveness of the category, likely due to the elimination of clutter on the shelf. Cachon et al. (2019) obtain a similar finding in the spirit of assortment attractiveness in U.S. automobile sales—when additional inventory expands the available set of submodels, sales increase; however, when additional inventory is of the same submodel as an existing vehicle, sales decrease. In our setting, because inventory is held in the back room, product display on the shop floor is not affected by the stockouts of some SKUs, and consumers observe the same assortment of designs and colors. Therefore, the assortment attractiveness in a store does not change, and the positive or negative assortment attractiveness effects of stockouts, as observed by Kalyanam et al. (2007), do not exist in our setting.

Third, most of the MNL choice-model based research studies use data from grocery retail stores and find that stockouts increase the sales of other items through the demand spillover effect. Presumably this spillover effect will exist in fashion retail settings such as ours. Specifically, when a specific size of a product is out of stock, the unmet demand will spill over to adjacent sizes. We make the following assumption on what adjacent sizes will be affected by this stockout-based demand spillover effect.

Assumption (Cross-Size Demand Spillover). In the footwear retail setting, when a product's size-s SKU is out of stock, its unmet demand will spill over to adjacent sizes (i.e., $s+0.5$ or $s-0.5$) of the same product but will not spill over to sizes larger than $s+0.5$ or smaller than $s-0.5$ (assuming the U.S. shoe size system with 0.5 size interval).

This assumption is reasonable because sports shoes with a too large or too small size are most likely unfit for a consumer. Therefore, the stockout of size-s will cause no or negligible demand spillovers to sizes larger than $s+0.5$ or smaller than $s-0.5$.

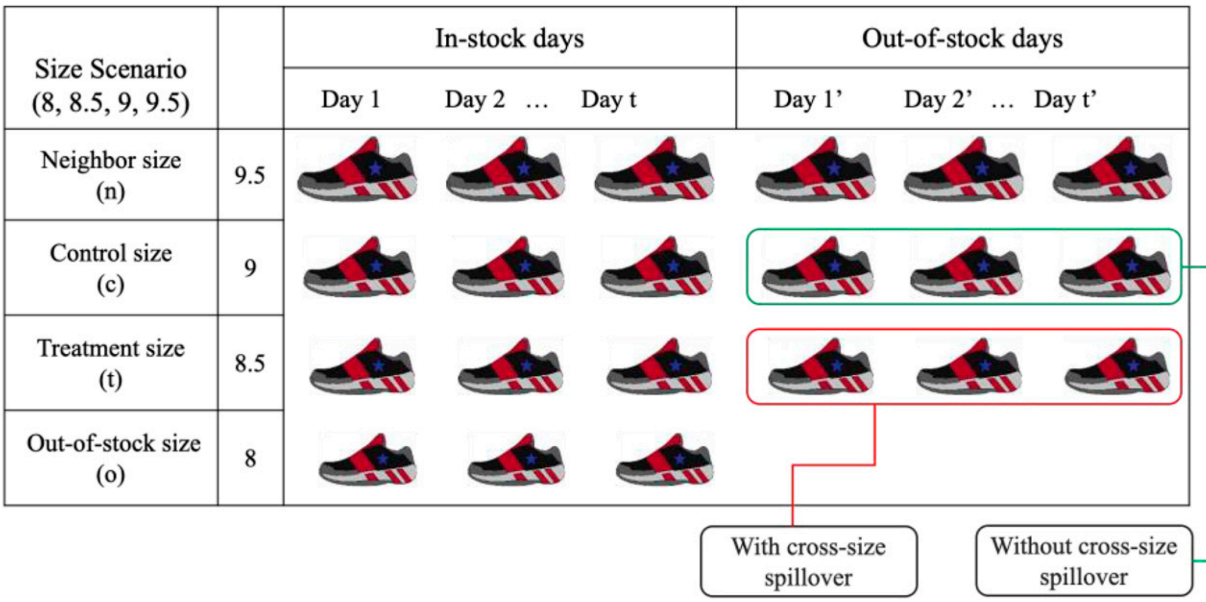
In many retail settings, it is difficult to isolate each of the aforementioned three effects of inventory stockouts because they are entangled with each other. Fortunately, in our footwear retail setting, the store layout with the shop floor displaying samples and the back room holding inventory allows us to eliminate the effects of "broken assortment" and "assortment attractiveness." Therefore, our empirical setting allows us to isolate the demand spillover effect of stockouts.

4.3. DID Estimation Framework

To estimate the cross-size demand spillover effect as described in Section 4.2, we develop a DID estimation framework which leverages the granularity of our data set. The key idea is that when the stockout of an SKU occurs, only some SKUs will experience the cross-size spillover effect, whereas others remain unaffected. Hence, we can view an SKU's stockout event as a treatment.

Figure 3 illustrates our DID framework to estimate the cross-size spillover effect of stockouts. As an example, let us suppose a specific product's size-8 SKU is experiencing a stockout event. During the size-8 SKU's out-of-stock days, the daily sales of the same product's size-8.5 SKU will be affected by the cross-size demand spillover effect, whereas the same product's size-9 SKU will not be affected (recall the assumption in Section 4.2). Therefore, we place the size-8.5 SKU in the "treatment group" and place the size-9 SKU in the "control group." To prevent the treatment and control groups' daily sales from being affected by other stockout events, we choose a DID observation window spanning the size-8 SKU's in-stock and out-of-stock days such that the size-8.5 SKU (treatment size), the size-9 SKU (control size), and the size-9.5 SKU (neighbor size) are always in stock. The in-stock status of the size-9.5 SKU helps to rule out the potential spillover

Figure 3. (Color online) DID Estimation for the Cross-Size Spillover Effect: Size Scenario (8, 8.5, 9, 9.5)



effect from its stockout on the sales of the control size (i.e., size 9). Therefore, by constructing a sample containing the treatment and control groups' daily sales during the in-stock and out-of-stock days of the size-8 SKU, the cross-size demand spillover effect can be estimated via a DID approach. The detailed DID specification will be given in Section 5.2.

It should be noted that Figure 3 only depicts a specific scenario that captures the cross-size spillover effect from a smaller size to a larger size. In contrast, if we select size 9.5 as the out-of-stock size, size 9 as treatment size, size 8.5 as control size, and size 8 as neighbor size, and have the four SKUs' inventory statuses follow Figure A2 in Online Appendix A, then we have a scenario for which we can estimate the cross-size spillover effect from a larger size to smaller size. Our DID estimation will cover 10 different size scenarios as shown in Table 3.

Table 3. Size Scenarios

Scenario (o, t, c, n)	Out-of-stock size o	Treatment size t	Control size c	Neighbor size n
1	6.5	7	7.5	8
2	7	7.5	8	8.5
3	7.5	8	8.5	9
4	8	8.5	9	9.5
5	8.5	9	9.5	10
6	8	7.5	7	6.5
7	8.5	8	7.5	7
8	9	8.5	8	7.5
9	9.5	9	8.5	8
10	10	9.5	9	8.5

We shall comment on the validity of our DID specification. Given our data structure, the treatment and control groups naturally form well-matched pairs. The treatment SKU (size 8.5) and the control SKU (size 9) have almost all characteristics in common: they are sold in the same store and have identical product characteristics except size. Furthermore, during the in-stock days of the size-8 SKU, the treatment and control SKUs are not only in stock but also face identical store and market conditions. The only difference between them is the 0.5 size gap, which results in an exogenous sales difference between the two SKUs because of the natural foot-size distribution of a common consumer population shopping at the store. Consequently, the treatment SKU and the control SKU naturally have parallel sales pretrend. This claim will be validated by a pretrend test in Section 5.3. During the out-of-stock days of the size-8 SKU, the treatment and control SKUs are again not only in stock, but also face identical store and market conditions with the only exception that the stockout of the size-8 SKU leads to a spillover effect on the treatment SKU but not on the control SKU.

Comparing our DID framework to the estimation approaches in the existing literature, we see the following advantages. First, because of its simplicity, the DID framework does not suffer the curse of dimensionality. Anupindi et al. (1998, p. 414) point out that their estimation model in faces sample size problems. This is because for N products, "up to approximately N^2 spillover rates must be estimated for each of about 2^N possible states representing the various possible stockout conditions." Using attribute-based models to estimate stockout effects reduces dimensionality; however, it faces the practical challenge that fashion products' styles are difficult to be fully characterized by their attributes let alone represent them numerically (Ferreira et al. 2020). Second, certain combinations of stockouts may not occur in the data set. Hence, in many empirical settings, the estimates can be unreliable because of inadequate sample sizes. A common way to overcome this problem is to place "appropriate restrictions on substitution effects" (Anupindi et al. 1998, p. 408) or to use expert opinions to rule out certain substitutions (Fisher and Vaidyanathan 2014). Our DID framework does not need to impose such restrictions but leverages the granularity of the data to estimate the cross-size spillover effect.

5. Estimating the Cross-Size Demand Spillover Effect

In Section 5.1, we describe the sample construction process. In Section 5.2, we specify the DID regression and present our main results. To demonstrate the validity of the estimation framework, we present a pretrend test in Sections 5.3, a placebo test in Section 5.4, and conduct robustness check in Section 5.5.

5.1. Sample Construction

To apply the DID estimation framework as described in Section 4.3, we construct a sample such that the inventory status of four consecutive sizes of a product has the structure illustrated in Figure 3. During "in-stock days" as illustrated in Figure 3, neither the treatment size nor the control size is affected by the cross-size spillover effect caused by the stockout of the out-of-stock size. During out-of-stock days, the treatment size is affected by the cross-size spillover effect, whereas the control size is not because it is beyond the size range of ± 0.5 from the out-of-stock size. Meanwhile, we make sure the "neighbor size" is always in stock during both in-stock days and out-of-stock days so that it does not impose a cross-size spillover effect on the control size. It should be noted that we use an SKU's inventory at the beginning of a day (00:01 AM) to determine its inventory status during that day. That is to say, if an SKU is in stock during part of the day and then gets sold out, we still set its inventory status for the day as in stock. Although this approach can cause inaccuracies in an SKU's inventory status in our sample, we can show that the percentage of time such an incident arises is negligible compared with an SKU's life cycle (Online Appendix D1), thereby having little effect on our estimation.

To proceed, we select the target stores, target footwear products, and target sizes from the raw data. For target stores, we select our retail partner's 217 stores, which are dedicated for distributing sportswear products of brand A. For brand A's footwear product family, the raw data consist of 5,258 men's and 2,000 women's sportswear products during the two-year sample period. We focus on men's footwear products because they account for more than 70% of all footwear products. We select 503 men's footwear products with sales rank above the 90th percentile. These products were sold in most of the retail stores in the data set and accounted for 52% of the total sales revenue of men's footwear products. For the target sizes, eight standard sizes of men's sports shoes are considered, which include 6.5, 7, 7.5, 8, 8.5, 9, 9.5, and 10.

According to Figure 3's data structure, 10 different size scenarios can be formed given the eight standard sizes from 6.5 to 10, as shown in Table 3. Each of the size scenarios is denoted by a four-element array (o, t, c, n), which stands for out-of-stock size, treatment size, control size, and neighbor size, respectively. Among the 10 size scenarios, Scenarios 1–5 have a cross-size spillover from a smaller size to a larger size, whereas Scenarios 6–10 have a cross-size spillover from a larger size to a smaller size.

Using 217 stores of brand A, 503 men's footwear products, and 10 size scenarios, we develop an algorithm to construct our DID sample. The main idea of the algorithm, as illustrated in Online Appendix C1, is to select SKU-store-day level observations with in-stock days and out-of-stock days in a 90-day time window, as illustrated in Step (4) of the algorithm. We use a 90-day time window because the average product life cycle in stores is around 90 days. Step (5) of the algorithm selects observations using 14 in-stock days and 14 out-of-stock days as cutoffs. We choose 14 days because on average it takes around two weeks to sell one unit of an SKU in a store according to the daily sales rates presented in Table 2. This step ensures that our constructed DID sample contains enough sale transactions to allow for estimation of the demand spillover effect. Other time cutoffs are considered as robustness check in Section 5.5. Applying this algorithm to our raw data, we obtain a sample with 4,024 SKUs and 1,496,970 observations. We define the variables used in our subsequent analysis and provide their summary statistics in Table 4.

5.2. DID Regression

We now provide the specification for the DID regression. We assume that the sales of the size-s SKU of product j in store m on day d, denoted by Sale_{mjsd}, follow a Poisson process with rate λ_{mjsd}:

$$\text{Sale}_{mjsd} \sim Poisson(\lambda_{mjsd}).$$

Poisson distributions have been commonly used in the literature to model demand in fashion retail settings (Kalyanam et al. 2007, Caro and Gallien 2010).

Table 4. Summary Statistics

Variable	Definition	Observations	Mean	Standard deviation
$Sale_{mjsd}$	Daily sales of product j's size-s SKU in store m on day d	1,496,970	0.057	0.252
$Treat_{mjos}$	If in store m, product j's size o SKU is out-of-stock and product j's size-s SKU is in the treatment group (i.e., $s = o \pm 0.5$), it equals 1. If product j's size-s SKU is in the control group (i.e., $s = o \pm 1$), it equals 0.	1,496,970	0.500	0.500
OOS_{mjod}	If in store m, product j's size-o SKU is out-of-stock on day d, it equals 1 and otherwise 0.	1,496,970	0.415	0.493
Age_{jd}	Number of days between day d and product j's launch date	1,496,970	76.9	60.1
$Price_{mjd}$	The price offered for product j in store m on day d	1,496,970	766	269

Our DID specification is given by Equation (1), which applies a log linear model on the Poisson rate:

$$\begin{aligned}\log(\lambda_{mjsd}) = {} & \beta_0 + \beta_1 \mathrm{Treat}_{mjos} \times \mathrm{OOS}_{mjod} + \beta_2 \mathrm{Treat}_{mjos} \\ & + \beta_3 \mathrm{OOS}_{mjod} + \beta_4 \mathrm{Age}_{jd} + \beta_5 \log(\mathrm{Price}_{mjd}) \\ & + \mathrm{Store}_m + \mathrm{Product}_j + \mathrm{Size}_s \\ & + \mathrm{WeekDayDummies}_d \\ & + \mathrm{HolidayDummies}_d \\ & + \mathrm{YearMonthDummies}_d + \varepsilon_{mjsd}. \end{aligned} \tag{1}$$

The detailed variable definitions are given in Table 4. Simply put, Treat_{mjos} is the binary variable that indicates whether the size-s SKU of product j in store m is in the treatment group of the size-o SKU of product j, that is, whether $s = o \pm 0.5$. OOS_{mjod} is a binary variable that equals one if the size-o SKU of product j in store m is out of stock on day d and equals zero otherwise. Age_{jd} is the age of product j by day d. Price_{mjd} is the price of product j in store m on day d, which is the same for all sizes of product j. The coefficient β_1 of the interaction term Treat × OOS captures the cross-size spillover effect induced by the stockout of the size-o SKU of product j in store m on day d.

In this DID specification, we carefully control product age and price at the product-store-day level. To control for time-invariant unobserved factors, we include store fixed effects Store_m, product fixed effects $\mathrm{Product}_j$, and size fixed effects Size_s. For time fixed effects, we include weekday dummies, holiday dummies, and year-month dummies. We include time fixed effects at the year-month level to control for any underlying sales trend that affects all stores and SKUs. This yields an estimable and reasonable model given the pattern of daily sales in the data—it has weekly periodic fluctuations and large increases during holidays, which are controlled by the weekday and holiday dummies, but it does not have a strong fluctuation from week to week. The error term is clustered at the store-product level.

Table 5 displays the estimation results. The coefficients of the interaction term Treat × OOS in the three columns are all positive and statistically significant. For size scenarios 1–5, Column (1) indicates that the stockout of an SKU on average caused a 23.0% increase ($e^{0.207} - 1 = 0.230$) in the daily sales of the adjacent-larger-size SKU of the same style. For size scenarios 6–10, Column (2) indicates that the stockout of an SKU on average caused a 19.8% increase ($e^{0.181} - 1 = 0.198$) in the daily sales of the adjacent-smaller-size SKU of the same style. For size scenarios 1–10 combined, Column (3) indicates that the stockout of an SKU on average caused a 21.9% increase ($e^{0.198} - 1 = 0.219$) in the daily sales rate of the adjacent-smaller-size *or* the adjacent-larger-size of the same style. In summary, sports footwear products have a strong cross-size spillover effect in this brick-and-mortar retail setting.

Table 5 also shows the estimated coefficients for Price_{mjd} and Age_{jd}. For size scenarios 1–10 combined, Column (3) shows that a 1% decrease in the product price was associated with a 2.61% increase in the daily sales of the SKU. It also shows that a one-day increase in the product age was associated with a 0.2% ($e^{0.002} - 1 = 0.002$) decrease in the daily sales of the product.

Table 5 reports the demand spillover effect in terms of the increase of the daily sales of adjacent sizes.

Table 5. Cross-Size Spillover Effect

	Scenario 1–5 (1)	Scenario 6–10 (2)	Scenario 1–10 (3)
Treat × OOS	0.207***	0.181***	0.198***
	(0.018)	(0.025)	(0.015)
log(Price)	−2.690***	−2.626***	−2.610***
	(0.177)	(0.201)	(0.155)
Age	−0.002*	−0.002	−0.002*
	(0.001)	(0.001)	(0.001)
Store fixed effects	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes
Size fixed effects	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes
Observations	923,844	573,126	1,496,970
Log likelihood	−197,485	−114,235	−312,478

Note. Standard errors clustered by store-product tuple.
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

To provide an intuitive interpretation of these results in terms of the stockout-based demand spillover effect, we estimate the fraction of an out-of-stock size's unmet demand that spills over to an adjacent size by using a slightly modified version of the DID specification given by Equation (1) earlier:

$$\begin{aligned}\log(\lambda_{mjsd}) = {} & \beta_0 + \alpha\left(\frac{\lambda_o}{\lambda_s}\right)\text{Treat}_{mjos} \times \text{OOS}_{mjod} + \beta_2\text{Treat}_{mjos} \\ & + \beta_3\text{OOS}_{mjod} + \beta_4\text{Age}_{jd} + \beta_5\log(\text{Price}_{mjd}) \\ & + \text{Store}_m + \text{Product}_j + \text{Size}_s \\ & + \text{WeekDayDummies}_d \\ & + \text{HolidayDummies}_d \\ & + \text{YearMonthDummies}_d + \varepsilon_{mjsd}, \qquad (2)\end{aligned}$$

where α is the coefficient representing the fraction of the out-of-stock size-o's unmet demand that spills over to an adjacent size s, and λ_0/λ_s is the demand ratio between size o and size s. λ_o/λ_s is assumed to be a constant, which captures the underlying demand difference between the two adjacent sizes. We can show that the DID coefficient, representing the relative sales increase of the treated size, is equal to $\alpha(\lambda_o/\lambda_s)$. A detailed derivation of this formula is provided in Online Appendix D2.

To proceed, we estimate the retailer's total sales of each size during the two-year sample period for size 6.5, 7, 7.5, 8, 8.5, 9, 9.5, and 10, respectively. Using these total sales to approximate the demand ratio λ_o/λ_s between two adjacent sizes, we run the DID regression specified by Equation (2) and report the results in Table 6. Using the coefficient of (λ_o/λ_s) Treat × OOS in Table 6, we estimate that on average 25.1% of the unmet demand of an out-of-stock SKU spills over to an adjacent-larger size and 26.6% spills over to an adjacent-smaller size. In total, the cross-size demand spillover effect captures 51.7% of the unmet demand of an out-of-stock SKU. We will refer to these percentages as the *theoretical* cross-size demand spillover effect because they are estimated using a constructed sample in which the out-of-stock SKU's adjacent sizes are always in stock. In Section 6.3, we will estimate the *actual* demand spillover effect, which takes into account the probability that the adjacent sizes are out of stock at the same time as the out-of-stock SKU, thereby reducing the observed cross-size spillover effect in terms of their sales increase.

Table 6. Cross-Size Spillover Effect: Fraction of Unmet Demand Spills over to Adjacent Sizes

	Scenario 1–5 (1)	Scenario 6–10 (2)	Scenario 1–10 (3)
(λ_o/λ_s) Treat × OOS	0.251***	0.266***	0.251***
	(0.020)	(0.033)	(0.018)
log(Price)	−2.690***	−2.627***	−2.611***
	(0.177)	(0.201)	(0.155)
Age	−0.002*	−0.002	−0.002*
	(0.001)	(0.001)	(0.001)
Store fixed effects	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes
Size fixed effects	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes
Observations	923,844	573,126	1,496,970
Log likelihood	−197,485	−114,235	−312,478

Note. Standard errors clustered by store-product tuple.
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

For boundary sizes such as size 6.5 and size 10 (the retailer rarely carries any size smaller than 6.5 or larger than 10 for men's footwear products), consumers only have the option to consider a single adjacent size: the only alternative is size 7 if size 6.5 is out of stock, whereas the only alternative is size 9.5 if size 10 is out of stock. One might be wondering whether the cross-size spillover effect might be larger for these boundary sizes. To answer this question, we conduct the regression specified by Equation (2) using a subsample consisting of only size Scenarios 1 and 10 (as shown in Table 3). The result is reported in Table B1 in Online Appendix B. The DID coefficient indicates that roughly 24.4% of unmet demand of a boundary size spills over to a neighboring size, which is similar to the percentages estimated for the interior sizes in Table 6. Based on this observation, we conjecture that the magnitude of cross-size demand spillover is likely driven by the distribution of foot sizes among consumers—given a pool of consumers with their most desired size being size o of the product, only the consumers with foot size on the larger (smaller) end of the pool will settle for size $o + 0.5$ (size $o - 0.5$).

5.3. Pretrend Test

To demonstrate the validity of our DID framework, we conduct a test to show that there does not exist different pretrends in the daily sales of the treatment and control groups. If a difference in the pretrends existed, then the cross-size spillover effect we find earlier could have been because of the difference in the sale trends of the two groups before treatment (e.g., the daily sales of the treatment group was already higher than that of the control group before the stockout event). To proceed, we adopt a pretrend test approach similar to the one used by Gallino and Moreno (2014) (see equation (2) on p. 1439 of the paper). For a specific size scenario (o, t, c, n) of product j in store m, the size-t SKU is in the treatment group, whereas the size-c SKU is in the control group. For this pair of SKUs, we only consider the in-stock days from the constructed sample; in other words, we select the pretrend observations of the treatment and control groups by removing those observations with $\text{OOS}_{mjod} = 1$ from the full sample constructed in Section 5.1. We

Table 7. Pretrend Test

	Scenario 1–5 (1)	Scenario 6–10 (2)	Scenario 1–10 (3)
Treat × Age	0.0002	−0.0001	0.0001
	(0.0002)	(0.0003)	(0.0002)
log(Price)	−2.705***	−2.644***	−2.608***
	(0.221)	(0.216)	(0.188)
Age	−0.002	−0.001	−0.002
	(0.001)	(0.002)	(0.001)
Store fixed effects	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes
Size fixed effects	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes
Observations	539,106	336,334	875,440
Log likelihood	−111,775	−65,686	−178,084

Note. Standard errors clustered by store-product tuple.
$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

then use the following Poisson regression model to perform a pretrend test.

$$\begin{aligned}\log(\lambda_{mjsd}) = {} & \beta_0 + \beta_1 \text{Treat}_{mjos} \times \text{Age}_{jd} + \beta_2 \text{Treat}_{mjos} \\ & + \beta_3 \text{Age}_{jd} + \beta_4 \log(\text{Price}_{mjd}) \\ & + \text{Store}_m + \text{Product}_j + \text{Size}_s \\ & + \text{WeekDayDummies}_d \\ & + \text{HolidayDummies}_d \\ & + \text{YearMonthDummies}_d + \varepsilon_{mjsd} \end{aligned} \tag{3}$$

In this specification, we again assume that the daily sales of the size-s SKU of product j in store m follows a Poisson process with rate λ_{mjsd}. The definitions of all the variables remain the same as in Equation (1). The standard errors are clustered at the store-product level.

β_1 is the coefficient of Treat × Age; thus, it captures the time-trend difference in the daily sales of the treatment and control groups. Table 7 presents the regression results. The results in Columns (1)–(3) are consistent: The coefficient of the Treat × Age is close to zero and statistically insignificant, suggesting that there do not exist statistically significant different sales pretrends between the treat and control groups. Therefore, we rule out the possibility that different pretrends drive the results of our DID estimation.

5.4. Placebo Test

In this section, we design a placebo test to validate our main DID estimation results. The placebo test begins with constructing a new sample by slightly modifying the sample construction process illustrated in Section 5.1. The essential idea is to select a larger size-gap between a fake "treatment" size and an out-of-stock size. Specifically, we will make the size gap greater than 0.5 so that the cross-size spillover effect should be close to zero and statistically insignificant. In other words, using this new sample with fake "treatment group" to conduct the DID regression, we shall not see any cross-size spillover effect because the larger size-gap make it infeasible for the cross-size demand spillover between the fake "treatment" size and the out-of-stock size.

For the placebo test, instead of using 4 sizes to define a size scenario (as shown in Table 3), we use five consecutive sizes to specify a size scenario and use a five-element array (o, l, t, c, r) to denote a size scenario, and the letters in the array stand for (out-of-stock, left neighbor, treatment, control, right neighbor). There are eight size scenarios as illustrated in Table B2 in Online Appendix B. To understand the size scenarios, let us consider an example: (6.5, 7, 7.5, 8, 8.5). In this size scenario, 6.5 is the out-of-stock size, whereas 7.5 is the fake "treatment" size. Due to the big size gap between 6.5 and 7.5, which equals 1 instead of 0.5, cross-size demand spillover between 6.5 and 7.5 is infeasible. In this size scenario, 8 is the control size, whereas both 7 and 8.5 are the neighbor sizes. We make sure both these neighbor sizes are always in-stock during the in-stock and out-of-stock days of size 6.5 so that they will not cause any cross-size spillover to the fake "treatment" size and the control size.

Using this new sample, we estimate the cross-size spillover effect using Equation (1). Table 8 presents the results of the placebo test. The results in Columns (1)–(3) are consistent: all of them show a close-to-zero and statistically insignificant coefficient of the interaction term Treat × OOS, which indicates that the cross-size spillover effect is insignificant, confirming our conjecture that when the size gap between the out-of-stock size and the treatment size is large, cross-size demand spillover is infeasible. This strongly supports that the estimated cross-size spillover effect is not due to unobserved trends or random noises in the daily

Table 8. Placebo Test

	Scenario 1–4 (1)	Scenario 5–8 (2)	Scenario 1–8 (3)
Treat × OOS	0.008	−0.024	−0.003
	(0.020)	(0.030)	(0.017)
log(Price)	−2.624***	−2.743***	−2.625***
	(0.214)	(0.198)	(0.191)
Age	−0.002	−0.002	−0.002*
	(0.001)	(0.001)	(0.001)
Store fixed effects	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes
Size fixed effects	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes
Observations	658,130	392,306	1,050,436
Log likelihood	−138,152	−83,225	−221,906

Note. Standard errors clustered by store-product tuple.
$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

sales of the treatment and control groups but caused by stockout-based demand spillovers between adjacent sizes.

5.5. Robustness Check

To further demonstrate the validity of our estimation results, we conduct regression analysis using two alternative specifications and conduct one sensitivity analysis of the sample construction process. In the first alternative specification, we replace the Poisson regression in our DID specification with zero-inflated negative binomial regression to address the issue of a large number of zero dependent variables in our data (SKUs' low daily demand rate leads to frequent zero daily sales at the SKU-store level). In the second alternative specification, we add SKU inventory levels to our DID regression to control for sales associates' reactions to inventory levels that might affect sales.[4] In the sensitivity analysis, we focus on the choice of 14-day cutoff in the sample construction process and estimate the DID regression under a series of cutoff days (i.e., 10, 12, 16, and 18 days). All three robustness checks produce highly similar estimation results as the baseline results reported in Section 5.2, thereby validating our main empirical findings. The details and the results of the three robustness checks are provided in Online Appendix E.

6. Discussions on Consumer Response to Inventory Stockouts

6.1. Heterogeneous Analysis: Cross-Size Demand Spillover Effect

Given the strong cross-size demand spillover effect demonstrated in Section 5, we are interested in exploring whether this effect varies across different types of stores and different product categories. Specifically, we compare the cross-size demand spillover effect between core stores and regular stores, high-price products and low-price products, casual sports shoes, and specialized sports shoes. To proceed, we conduct DID regressions with a three-way interaction term as follows:

$$\begin{aligned}\log(\lambda_{mjsd}) = {} & \beta_0 + \beta_1 \text{Treat}_{mjos} \times \text{OOS}_{mjod} \times \text{HTerm} \\ & + \beta_2 \text{Treat}_{mjos} \times \text{OOS}_{mjod} + \beta_3 \text{Treat}_{mjos} \\ & + \beta_4 \text{Treat}_{mjos} \times \text{HTerm} + \beta_5 \text{OOS}_{mjod} \\ & + \beta_6 \text{OOS}_{mjod} \times \text{HTerm} + \beta_7 \text{Age}_{jd} \\ & + \beta_8 \log(\text{Price}_{mjd}) + \text{Store}_m + \text{Product}_j \\ & + \text{Size}_s + \text{WeekDayDummies}_d \\ & + \text{HolidayDummies}_d \\ & + \text{YearMonthDummies}_d + \varepsilon_{mjsd}. \end{aligned} \tag{4}$$

In Equation (4), *HTerm* represents one of the heterogeneity dummy variables that we will define later. All other variables in Equation (4) remain the same as in Equation (1). The coefficient β_1 of the three-way interaction term $\text{Treat}_{mjos} \times \text{OOS}_{mjod} \times \text{HTerm}$ represents the marginal cross-size demand spillover effect because of the heterogeneity specified by HTerm. To proceed, we define three dummy variables to represent heterogeneous characteristics as follows.

First, we use IsCorestore_m to indicate whether store m is a core store or a regular store. Core stores play a more important role in our retail partner's business, generally are bigger, and carry a larger product assortment than regular stores. These core stores are either flagship stores or are located in prominent commercial districts. When an out-of-stock event occurs, consumers in core stores have more alternatives to buy and are more accessible to competitors' stores to buy the same brand or substitute brands. Therefore, we conjecture that the cross-size spillover effect in a core store is weaker than that in a regular store. Table 9 displays the regression results using the sample with all size scenarios. Specifically, Column (1) indicates that the cross-size spillover effect in a core store is 67.8% $(= (e^{0.229-0.068} - 1)/(e^{0.229} - 1))$ of that in a regular store, which confirms our conjecture.

Second, we use IsCasual_j to indicate whether product j is a "casual sports shoe" rather than a "specialized sports shoe" (basketball shoe or running shoe). While casual sports shoes are good for everyday use such as walking, they might not be suitable for intensive sports activities such as playing basketballs or running. Therefore, consumers buying specialized sports shoes might demand a higher level of size fit. Based on this logic, we conjecture that the cross-size spillover effect of casual sports shoes is stronger than that of specialized sports shoes. Table 9, Column (2), indicates that the cross-size spillover effect of casual sports shoes is 1.475 $(= (e^{0.160+0.068} - 1)/(e^{0.160} - 1))$ times of that of specialized sports shoes, which confirms our conjecture.

Third, we use IsHighprice_j to indicate whether product j is a high-price or low-price product using the median *original* price of all men's sports footwear products as the cutoff. The original price is set when a product is introduced to the market, and this pricing decision is independent of the inventory status of SKUs. Because consumers paying for an expensive product might demand a more stringent size fit, we conjecture that the cross-size spillover effect of high-price products is weaker than that of low-price products. Table 9, Column (3), indicates that the cross-size spillover effect of high-price products is 72.3% $(= (e^{0.223-0.057} - 1)/(e^{0.223} - 1))$ of that of low-price products, which confirms our conjecture.

Fourth, we use IsBigcity_m to indicate whether store m is located in a big city. We classify the different cities in our sample into big cities and small cities based on average annual income of consumers. In

Table 9. Heterogeneous Analysis

HTerm	*IsCorestore* (1)	*IsCasual* (2)	*IsHighprice* (3)	*IsBigcity* (4)	*IsNearbyAvail_2km* (5)	*IsNearbyAvail_3km* (6)
Treat × OOS × HTerm	−0.068*	0.068*	−0.057*	−0.004	−0.012	−0.015
	(0.027)	(0.029)	(0.028)	(0.027)	(0.028)	(0.023)
Treat × OOS	0.229***	0.160***	0.223***	0.199***	0.197***	0.201***
	(0.019)	(0.022)	(0.019)	(0.020)	(0.018)	(0.020)
log(Price)	−2.610***	−2.614***	−2.612***	−2.610***	−2.610***	−2.610***
	(0.155)	(0.155)	(0.154)	(0.155)	(0.155)	(0.155)
Age	−0.002*	−0.002*	−0.002*	−0.002*	−0.002*	−0.002*
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Store fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Size fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,496,970	1,496,970	1,496,970	1,496,970	1,496,970	1,496,970
Log likelihood	−312,459	−312,432	−312,450	−312,459	−312,469	−312,471

Note. Standard errors clustered by store-product tuple.
$^*p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

cities like Shanghai (the largest city in China), there are more options for consumers to purchase from a competitor's store to get their desired sizes. Therefore, we conjecture that the cross-size spillover effect in big cities is weaker than that in small cities. However, Table 9, Column (4), indicates there does not exist a statistically significant difference between stores located in big cities and those located in small cities in terms of the cross-size spillover effect.

Finally, we explore whether the out-of-stock SKU is available in nearby stores moderates the cross-size spillover effect. One may conjecture that consumers will be less willing to settle for an adjacent size if his desired size is available in a nearby store. To test this, we use the longitude and latitude coordinates of each store to calculate store-to-store distances and define two binary variables, $\text{IsNearbyAvail}^{2\text{km}}_{\text{mjsd}}$ and $\text{IsNearbyAvail}^{3\text{km}}_{\text{mjsd}}$, which denote whether store m's nearby stores (within a 2-km and 3-km range, respectively) have product j's out-of-stock size s available. Table 9, Columns (5) and (6), show that the impact of nearby stores' availability on the cross-size spillover effect is statistically insignificant.

6.2. Estimating Cross-Style Demand Spillover Effect

In Section 5, we presented a DID framework to estimate the cross-size spillover effect of stockouts. When a consumer's desired size of a particular style is out of stock, our analysis thus far has shown that he has a strong tendency to purchase an adjacent size of the same style. Nevertheless, the consumer might also purchase different styles of his desired size. In this section, we will empirically estimate the cross-style demand spillover effect at the SKU level. The DID framework we presented in Section 5 cannot be easily adapted to estimate the cross-style spillover effect mainly because it is difficult to identify suitable SKUs as the treatment and control groups for a specific SKU stockout event. This is because there are a large number of styles being sold in a store at the same time and frequent stockouts happen to many styles, both of which makes it challenging to accurately estimate the cross-style spillover effect.

Based on our literature review (see Section 2 for details), there does not seem to exist a "perfect" method for estimating the cross-style spillover effect for fashion products. The large number of SKUs in our setting prevents us from using product-substitution models because of the curse of dimensionality. To reduce the dimensionality, we resort to attribute-based substitution models. Specifically, we adapt the demand estimation model of Fisher and Vaidyanathan (2014) to our setting to estimate the cross-style spillover effect. Fisher and Vaidyanathan (2014) develop an attribute-based demand estimation model that captures substitution probabilities across product attribute levels. In this subsection, we first adapt this demand model to our setting of sports footwear products. Then we introduce the product attributes that are used in the estimation. This section ends with the estimation results and the discussion about this attribute-based demand estimation model in the fashion retail setting.

6.2.1. Adapted Demand Estimation Model of Fisher and Vaidyanathan (2014).

Fisher and Vaidyanathan (2014) develop an attribute-based model to estimate the product substitution patterns within a store. For simplicity, we drop the store index and provide a brief description about the model and the estimation procedure. First, an SKU is viewed as a collection of attributes.[5] Let A denote the number of attributes, let a denote the attribute type index, and let N_a denote the number of levels of attribute $a \in \{1, 2..., A\}$; f_{au} is the fraction of consumers who prefer level $u \in \{1, 2, ., N_a\}$ for attribute a; and π_{auv} is the substitution probability that a consumer, whose first choice on attribute a is level u, is willing to substitute v for u. Second, the demand model and substitution patterns are defined as follows. For SKU i with attributes $i_1, i_2, \ldots, i_A$, the fraction of consumers who most prefer this SKU is assumed to be $f_i = \prod_{a=1}^{a=A} f_{ai_a}$, which is the demand share for the SKU. Let S_d denote a store's assortment on day d. If SKU i is available in S_d, a consumer who prefers i will buy it. Otherwise, this consumer will buy SKU j instead (where $j = j(i, S_d) = \arg\max_{j \in S_d} \prod_{a=1}^{a=A} \pi_{ai_a j_a}$) with probability $\pi_{ij} = \prod_{a=1}^{a=A} \pi_{ai_a j_a}$. Finally, the likelihood function is constructed as follows. Based on the demand model, a consumer faced with assortment S_d purchases SKU j with probability $F_j(S_d) = f_j + \sum_{i \notin S_d, j(i,S_d)} f_i \pi_{ij}$, and the probability that a consumer makes any purchase is $F(S_d) = \sum_{j \in S_d} F_j(S_d)$. Given the sale x_{jd} of SKU j on day d and assortment S_d, the log-likelihood function across all days in the sample period is

$$LLH(f, \pi) = \sum_d \left[\sum_{j \in S_d} x_{jd} \log F_j(S_d) - \left(\sum_{j \in S_d} x_{j_d} \right) \log F(S_d) \right]. \tag{5}$$

This log-likelihood function considers the fact that in our setting store assortment S_d may change over the sample period, which differs from the static assortment setting of Fisher and Vaidyanathan (2014). Parameters (f, π) can be estimated by maximizing this function. Online Appendix F contains more details of the demand model and a complete formulation of the maximization problem.

6.2.2. Attributes and Attribute Levels of Men's Sports Footwear Products. To apply the demand estimation procedure of Fisher and Vaidyanathan (2014), we need to describe an SKU by a set of attributes and attribute levels. Therefore, we need to discretize all available attributes in the data set. However, we shall point out that the available attributes in the data set cannot completely specify the style of a footwear product. For example, the color pattern of a particular footwear product cannot be simply described by a single color because most sports shoes have at least two colors, which occupy different surface areas of the shoes. Machine learning–based methods have been developed to tackle the challenge of digitizing the color patterns of fashion products. For example, Li et. al. (2020) develop a process to extract color information from shoe images and then reconstruct the color attribute in a relatively low dimensional space. Besides colors, the design of a footwear product cannot be completely described by the limited attributes available in the data set. The challenge of translating the style of a product into numerical data has been noted in the most recent demand estimation literature (Ferreira et al. 2020).

Given the limitation of the data set in providing the necessary attributes to completely specify the style of a product, we focus on the following three major attributes: color, shape, and price. Using the color bar code in product ID, we extract the base color for each product and group them into four levels: black, white, cool color, and warm color. The shape attribute is classified into two levels: medium top and low top. We discretize the price attribute to five tiers (tier 1, tier 2, … , tier 5) to represent five different levels with one as the lowest and five as the highest. These price tiers reflect the style and quality of products and are fixed; that is, they are not affected by price promotions and inventory status. In summary, we estimate this demand model using four colors, two shapes, and five price tiers. These relatively small number of attributes and attribute levels allow us to keep computational complexity at a reasonable level.

6.2.3. Estimation. As an illustration of the estimation procedure described in Section 6.2.1, we construct a sample using the top 10 stores (based on annual sales volumes) and size 8 casual sports shoes for the following reasons. First, selecting the top 10 stores ensures sufficient sale transactions for an accurate estimation. Second, size 8 is one of the most popular sizes and thus has sufficient sale transactions to allow estimation at the store level. Third, we can focus on the casual sports shoe category because cross-category demand spillover is generally infeasible (e.g., unmet demand of basketball shoes generally does not spill over to casual shoes). Given the four colors, two shapes, and five price tiers as explained in Section 6.2.2, there are 40 possible distinct SKUs. Based on this attribute definition, the average number of daily available SKUs ranges from 14.7 to 22.8 in the sample. The total sales of all size 8 casual sports shoes vary between 1,164 and 5,529 over the two-year sample period. On average, the monthly sales are 2.3 pairs of shoes per SKU. Applying the demand model illustrated in Section 6.2.1 to this sample, we estimate the average values of the demand shares and attribute substitution probabilities (f, π), which are reported in Tables 10 and 11.

Table 10 displays the demand shares of different SKUs. It shows that black shoes and white shoes are generally more popular than shoes with other colors. It also shows that most consumers prefer casual sports shoes with a low top and that a product's demand share is decreasing in its price. Table 11 shows the substitution probabilities between different levels of colors, shapes, and price tiers. The substitution matrix of the color attribute in Table 11 implies that white is

Table 10. Cross-Style Spillover Effect: Estimated Demand Shares (Averaged Across Stores)

Color	Demand share	Shape	Demand share	Price	Demand share
Black	46.67%	Low top	73.51%	Tier 1	24.45%
Cool	17.97%	Medium top	26.49%	Tier 2	24.58%
Warm	16.15%			Tier 3	23.43%
White	19.21%			Tier 4	16.36%
				Tier 5	11.18%

more likely to serve as a substitute for other colors. The substitution matrix of the shape attribute in Table 11 implies that consumers have a low tendency to substitute between a high-top and a low-top shoe style. The substitution matrix of the price tier attribute implies that the spillover effect from a higher price tier to a lower price tier is on average stronger than the spillover effect in the opposite direction.

To compute the *theoretical* cross-style demand spillover effect, we use the estimated parameters (f, π) and assume all SKUs other than the out-of-stock SKU are in stock. Recall that S_d denotes a store's assortment on day d. For an out-of-stock SKU i on day d $(i \notin S_d)$, if we assume all SKUs other than i can be potential substitutes, then its demand f_i $(f_i = \prod_{a=1}^{a=A} f_{ai_a})$ should spill over to SKU j (where $j = j(i, S - \{i\}) = \arg\max_{j \in S-\{i\}} \prod_{a=1}^{a=A} \pi_{ai_a j_a})$ with probability $\pi_{ij} = \prod_{a=1}^{a=A} \pi_{ai_a j_a}$. Here S denotes the set of complete assortment, which includes all possible combinations of the attribute types and levels (i.e., the aforementioned 40 distinct SKUs). The average percentage of unmet demand of the out-of-stock SKU i that spills over to other styles is equal to

$$\frac{\sum_d \sum_{i \notin S_d} f_i \pi_{i,j(i,S-\{i\})}}{\sum_d \sum_{i \notin S_d} f_i}. \tag{6}$$

Applying this formula to each of the top 10 stores, we find that the percentage of unmet demand that spills over to different styles ranges from 19.9% to 20.4%, with the average being 20.2%. These percentages suggest that compared with the cross-size demand spillover effect, which captures 51.7% of the unmet demand, the magnitude of the cross-style spillover effect is much smaller.

To summarize, we adapted the demand model of Fisher and Vaidyanathan (2014) to estimate the cross-style spillover effect. The model significantly reduces the dimension of the parameter space and makes it feasible to estimate the demand share and substitution probabilities at the store level. Nevertheless, as we pointed out earlier in Section 6.2.2, the attributes of fashion products cannot be easily digitized, nor can they be completely defined using the current data. Therefore, we caution that the estimated cross-style spillover effect is an approximation of the more granular stockout-based cross-style demand spillover patterns between SKUs in reality. We hope our study draws more research attention to this area as more granular project attributes data from the fashion retail industry are becoming available to researchers.

6.3. Managerial Insight: Stockout-Based Demand Spillovers and Lost Sales

To obtain a complete picture of consumer response to inventory stockouts in this fashion retail setting, we estimate the percentage of unmet demand of an out-of-stock SKU that spills over to different sizes, spills over to different styles, or becomes lost sales. To provide a realistic estimate of lost sales, we introduce the concept of the *actual* demand spillover effect and contrast it with the *theoretical* demand spillover effect, which has been estimated in Section 5.2 (cross-size)

Table 11. Cross-Style Spillover Effect: Estimated Substitution Probabilities (Averaged Across Stores)

Color	Black	Cool	Warm	White	
Black	1.000	0.083	0.114	0.238	
Cool	0.062	1.000	0.140	0.152	
Warm	0.049	0.078	1.000	0.208	
White	0.073	0.103	0.155	1.000	
Shape	**Low top**	**Medium top**			
Low top	1.000	0.151			
Medium top	0.116	1.000			
Price	**Tier 1**	**Tier 2**	**Tier 3**	**Tier 4**	**Tier 5**
Tier 1	1.000	0.070	0.122	0.070	0.089
Tier 2	0.098	1.000	0.123	0.147	0.094
Tier 3	0.139	0.080	1.000	0.071	0.119
Tier 4	0.119	0.134	0.186	1.000	0.048
Tier 5	0.142	0.166	0.174	0.243	1.000

and in Section 6.2.3 (cross-style). Theoretical spillover is the demand spillover that can occur when SKUs other than the out-of-stock SKU are in stock, whereas actual spillover is the demand spillover observed in stores which depends on the stores' real product assortment and inventory status.

In Section 5.2, we use the DID approach to estimate the theoretical cross-size demand spillover effect—roughly 25.1% (26.6%) of an out-of-stock SKU's unmet demand spills over to an adjacent-larger (adjacent-smaller) size of the same style. In total, 51.7% of an SKU's unmet demand spills over to adjacent sizes. This estimation is obtained using a constructed sample in which the adjacent sizes of the out-of-stock SKU are always *in stock*. In reality, an out-of-stock SKU's adjacent sizes can also be out of stock at the same time. Thus, the percentage of unmet demand that is actually captured through cross-size spillovers will be smaller than 51.7%.

To illustrate, we consider the constructed sample used in Section 6.2.3 that involves top 10 stores' size 8 casual shoes. For the focal size 8, there are in total 28,331 stockout events at the SKU-store-day level. Among these events, 9,515 are accompanied by stockout events of the size 8.5 SKU of the same style, and 15,682 are accompanied by stockout events of the size-7.5 SKU of the same style. That is to say, when a size-8 SKU is out of stock, the conditional probability of its adjacent-larger size also being out of stock is approximately 33.6% (=9,515/28,331), and the conditional probability for the SKU's adjacent-smaller size (size 7.5) being out of stock is approximately 55.4% (=15,682/28,331). As a result, the percentage of unmet demand that can be captured by its adjacent-larger size SKU is 25.1% × (1% − 33.6%) = 16.7%, whereas the percentage of unmet demand that can be captured by its adjacent-smaller size SKU is 26.6% × (1% − 55.4%) = 11.9%. In total, the *actual* cross-size spillover effect is 28.6% (=16.7% + 11.9%), which is much smaller than 51.7%, the *theoretical* cross-size spillover percentage.

In Section 6.2.3, we estimated the theoretical cross-style demand spillover effect— roughly 20.2% of the unmet demand of an out-of-stock SKU spills over to a different style when all other SKUs are assumed to be in stock. To estimate the actual cross-style demand spillover effect, we use a store's daily assortment S_d. If some SKUs other than the focal out-of-stock SKU are also out of stock, then the magnitude of the observed cross-style demand spillover will be lower. To illustrate, we modify the calculation process in Section 6.2.3 using the following formula:

$$\frac{\sum_d \sum_{i \notin S_d} f_i \pi_{i,j(i,S_d)}}{\sum_d \sum_{i \notin S_d} f_i}. \tag{7}$$

The difference between this formula and Equation (6) in Section 6.2.3 is that for an out-of-stock SKU i on day d ($i \notin S_d$), only the SKUs that are in the store's assortment S_d on day d (instead of S, which is the complete assortment) can be potential substitutes. Applying this new formula to the sample of top 10 stores constructed in Section 6.2.3, we show that the cross-style spillover percentage ranges from 13.2% to 16.1% across stores, with an average being 14.9%, which represents the *actual* cross-style spillover effect and is smaller than 20.2%, the *theoretical* cross-style spillover percentage.

To summarize, we have shown that in this retail setting, the *actual* percentages of stockout-induced unmet demand that are captured by cross-size and cross-style spillovers are 28.6% and 14.9%, respectively. Therefore, the remaining 56.5% of the unmet demand becomes lost sales. By contrast, if the store carries a complete assortment and has all other SKUs in stock, the *theoretical* cross-size and cross-style spillover percentages are 51.7% and 20.2%, respectively, and the remaining 28.1% of the unmet demand becomes lost sales. The significant reduction in lost sales suggests that capturing demand spillovers is a key mechanism for why carrying inventory mitigates the negative effect of stockouts on sales.

7. Managerial Implications on Interstore Proactive Transshipment

We explore the managerial implications of the cross-size demand spillover effect in the context of an interstore proactive transshipment problem. Our retail partner does not manufacture but purchases brand A products from the brand's original manufacturer. Because of long lead time, the retailer is unable to have in-season replenishment and relies on proactive transshipment to mitigate stockout-induced lost sales. *Proactive* transshipment is to reallocate inventory among stores *before* stockouts occur (Paterson et al. 2011). This is different from *reactive* transshipment that transfers inventory between stores *after* stockouts occur. The retailer makes transshipment decisions at the corporate level and had been struggling with excessively high transshipment costs, which prompted the collaboration with us to improve its proactive transshipment policy.

Given our empirical findings earlier, a proactive transshipment policy that ignores the cross-size spillover effect will lead to unnecessary transshipment and reduce profitability. To address this issue, we conduct a simulation analysis to study a single-period, multisize product's proactive transshipment problem. We build an optimization model and compare two solutions: one ignores the cross-size demand spillover effect, whereas the other does not. For a given product, there are K consecutive sizes sold in M stores. At T days before the end of the product life cycle, the retailer observes the inventory level I_{mk} of the size

$k \in \{1,2,\dots,K\}$ SKU in store $m \in \{1,2,\dots,M\}$. By this time, SKU inventory distributions in the stores are unbalanced because of different store-level demand realizations.

We make several assumptions to formulate the proactive transshipment optimization problem. First, we follow the literature by assuming that the primary daily demand of the focal product's size-k SKU in store m follows a Poisson process with rate λ_{mk} (Caro and Gallien 2010). Second, the total daily demand is the sum of the primary demand and the stockout-based cross-size demand spillover from adjacent sizes. For simplicity, the percentages of unmet demand that spills over to the adjacent-larger size and to the adjacent-smaller size are both assumed to be ρ. Third, the unit gross profit is a constant p, the salvage value of excess inventory at the end of the selling season is zero, and the unit transshipment cost is a constant c. Similar assumptions have been adopted in the transshipment literature; for example, linear transportation cost was adopted by Karmarkar and Patel (1977). The proactive transshipment optimization problem can be formulated as a nonlinear integer program as follows:

$$\vec{Y}_m^{\,*} = \operatorname{argmax} \sum_{m=1}^{M}\left[pS\left(\vec{Y}_m, \vec{\lambda}_m\right) - \frac{c}{2}\sum_{k=1}^{K}|Y_{mk} - I_{mk}|\right], \quad \forall m$$
$$s.t. \sum_{m}^{M} Y_{mk} = \sum_{m}^{M} I_{mk}, \quad \forall k,\ Y_{mk} \in \{0,1,2,\dots\}, \quad \forall m,k.$$

In the previous formulation, I_{mk} and Y_{mk} are the inventory levels of product j's size-k SKU in store m before and after transshipment, respectively; and $\vec{Y}_m = \{Y_{m1}, Y_{m2}, \dots, Y_{mK}\}$ and $\vec{\lambda}_m = \{\lambda_{m1}, \lambda_{m2}, \dots, \lambda_{mK}\}$ are the inventory and demand rate vectors of the entire size assortment. The decision variables are $\vec{Y}_m$, $\forall m$. The objective function maximizes the profit (i.e., the total gross profit subtracts the transshipment cost), and $S(\vec{Y}_m, \vec{\lambda}_m)$ denotes the expected sales of product j in store m during the T days given $\vec{Y}_m$ and $\vec{\lambda}_m$. The two constraints are the inventory balance constraint and the integer constraint of inventory levels. This optimization problem does not have a closed-form solution. To solve it numerically, we first ignore the cross-size spillover effect to establish a benchmark solution through a greedy algorithm that guarantees optimality. For the original problem with the cross-size spillover effect, we introduce a heuristic that applies the greedy algorithm to generate an initial solution and then uses a search algorithm to improve it (see Online Appendix C2 for details).

We focus on the retailer's 30 brand A stores in Shanghai, the largest city in China. As the empirical analysis earlier, we consider 275 men's casual sports shoes (these products have high cross-size spillover effect according to our analysis earlier), each of which has eight consecutive sizes. We set $T = 30$, which means the retailer makes its proactive transshipment decision 30 days before the end of the product life cycle. We track the inventory status at the SKU-store level and obtain the inventory distribution for each size, which is presented in Table B3 in Online Appendix B.

In this sample, the average daily sales rates across all product-store pairs are 0.046, 0.060, 0.049, 0.080, 0.077, 0.062, 0.056, and 0.039 for sizes 6.5, 7, 7.5, 8, 8.5, 9, 9.5, and 10, respectively. The primary demand rate of each of the 30 stores is set to equal the average daily sales rate multiplied by a uniform random variable over the support of [0.90, 1.10] to capture sales rate dispersion across stores. We set the cross-size spillover rate $\rho = 0.3$. The unit gross profit is $p = 200$ RMB (roughly 30 USD) and the unit transshipment cost $c \in \{30, 60, 90\}$ RMB. These profit and transshipment costs were estimated based on our communications with the retailer. The inventory distribution in Table B5 is used to generate each store's inventory level 30 days before the end of the product life cycle (this process is repeated 10 times to obtain 10 random samples). We apply our solution algorithms to the 30 different problem scenarios (three cost scenarios for each of the 10 random samples) and compute the average profits and transshipment costs, which are reported in Table 12. According to Table 12, when the cross-size spillover effect is incorporated in the proactive transshipment decision, the retailer's profit can increase by 0.1%, 0.5%, and 1.0% for $c \in \{30, 60, 90\}$, respectively. Meanwhile, the retailer's transshipment cost can decrease by 7.1%, 16.4%, and 24.9%, respectively. Considering the retail chain's 735 stores nationwide in 2016 and assuming each store conducts four proactive transshipment decisions in a year (one for each selling season), and each decision involves 100 products, then the potential profit gain can be 0.67, 3.07, and 5.27 million RMB, respectively, for the three cost scenarios.

Table 12. Simulation Analysis: A Single Product Sold in 30 Stores Located in Shanghai

Unit transshipment cost	Ignoring spillover		Considering spillover		Percent change	
	Profit	Transshipment cost	Profit	Transshipment cost	Profit	Transshipment cost
30	59,296	3,165	59,365	2,940	0.1%	−7.1%
60	56,685	4,902	56,998	4,098	0.6%	−16.4%
90	54,949	5,319	55,487	3,996	1.0%	−24.9%

8. Conclusion

This paper empirically investigates the stockout-based demand spillover effect in a brick-and-mortar fashion retail setting. Based on the real-time and granular sales and inventory records at the SKU-store-day level, we develop a difference-in-differences regression approach to estimate the stockout-based cross-size demand spillover effect. This method is deployed on a large-scale data set consisting of 217 stores, 503 products, 4,024 SKUs, and approximately 1.5 million observations. We further characterize the cross-size spillover effect's heterogeneity with respect to store type, product category, price, and location. Our estimation results show that an out-of-stock SKU's cross-size demand spillover effect is highly significant—51.7% of the unmet demand of an out-of-stock SKU spills over to adjacent sizes of the same style when they are in stock. As far as we know, our paper is the first to empirically demonstrate the strong stockout-based cross-size demand spillover effect in a fashion retail setting. By adapting the demand model of Fisher and Vaidyanathan (2014), we estimate that roughly 20.2% of the unmet demand of an out-of-stock SKU spills over to different styles when they are in stock. Taken together, these estimations suggest that about 28.1% of the unmet demand of an out-of-stock SKU becomes lost sales. We further find that when stockouts are widespread among SKUs, stockout-based demand spillovers are significantly reduced, resulting in much increased lost sales.

This paper makes two contributions. (1) It introduces an estimation framework to quantify the stockout-based cross-size demand spillover effect at the SKU level. This framework neither suffers the curse of dimensionality nor imposes strong assumptions on the spillover patterns. Therefore, it can be deployed to fashion retail settings with hundreds of stores and thousands of SKUs, which pose modeling and computational challenges to existing estimation frameworks. (2) Our paper generates novel insights into consumer response to inventory stockouts of fashion products, an understudied topic in the existing literature. Our simulation analysis demonstrates that incorporating the cross-size demand spillover effect into the retail chain's interstore proactive transshipment decision can reduce its transshipment cost substantially and improve its profitability.

Although we demonstrate our framework using footwear products, our estimation method can be applied to other fashion products with size assortment. If a fashion product's inventory is visible to consumers in the store, we can adapt our method to control for the "broken assortment" and the "assortment attractiveness" effects, which occur at the store or product category level. Our DID approach automatically selects the treatment and control SKUs in the same store and same product category. Adding time-variant assortment inventory levels into the DID regression will further control for the effects of dynamic product assortment on individual SKU sales.

Our study is not without limitations. Our estimated cross-size demand spillover effect might be partially due to the sales effort of sales associates, rather than completely due to consumers' true substitution willingness. The structure of our data does not allow us to isolate these two effects. However, from the retailer's perspective, the combined demand spillover effect is what matters to the company in terms of its profitability and inventory policy. To identify the behavioral mechanisms for the cross-size demand spillover effect, field or laboratory experiments can be conducted to further understand consumers' responses to inventory stockouts. This is a research question we plan to pursue in the future.

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Endnotes

[1] Caro and Gallien (2010) use Zara's data to construct a measure of stockout using the number of days when there was a stockout of a major size among all apparel articles in a store but do not present related statistics. Boada-Collado and Martínez-de-Albéniz (2020) find that in a fashion chain's 85 European stores, the stockout rate was extremely low at round 0.1% at the product-store-day level. Because in this study, a product is defined as "a distinct combination of model and color and may include several sizes," the stockout rate at the SKU (i.e., size-product-store-day) level is unclear.

[2] Because of data nondisclosure agreement, we cannot reveal the name of the brand.

[3] We excluded a rare size 10.5 from our study because it was only carried for some products occasionally at some stores and the number of observations were negligible compared with the eight standard sizes included in our sample.

[4] In Section 4.2, we assume that an SKU's sales rate is not affected by its inventory level because the inventory is stored in the backroom which cannot be observed by consumers. Nevertheless, sales associates can observe SKU inventory levels, which might affect their decisions on which products to recommend to consumers.

[5] The term SKU is used here to be consistent with Fisher and Vaidyanathan (2014). Nonetheless, it is not necessarily identical to a real SKU in stores because the attribute levels used in the estimation model might be less granular than the real product attributes to attain computational feasibility and tractability. See, for example, in the snack cake application in Fisher and Vaidyanathan (2014), different package sizes are grouped into two attribute levels: single serve and family size.

References

Anderson ET, Fitzsimons GJ, Simester D (2006) Measuring and mitigating the costs of stockouts. *Management Sci.* 52(11):1751–1763.

Anupindi R, Dada M, Gupta S (1998) Estimation of consumer demand with stock-out based substitution: An application to vending machine products. *Marketing Sci.* 17(4):406–423.

Boada-Collado P, Martínez-de-Albéniz V (2020) Estimating and optimizing the impact of inventory on consumer choices in a fashion retail setting. *Manufacturing Service Oper. Management* 22(3):582–597.

Cachon G, Gallino S, Olivares M (2019) Does adding inventory increase sales? Evidence of a scarcity effect in U.S. automobile dealerships. *Management Sci.* 65(4):1469–1485.

Campo K, Gijsbrechts E, Nisol P (2003) The impact of retailer stockouts on whether, how much, and what to buy. *Internat. J. Res. Marketing* 20(3):273–286.

Caro F, Gallien J (2010) Inventory management of a fast-fashion retail network. *Oper. Res.* 58(2):257–273.

Che H, Chen X, Chen Y (2012) Investigating effects of out-of-stock on consumer stockkeeping unit choice. *J. Marketing Res.* 49(4):502–513.

Conlon CT, Mortimer JH (2013) Demand estimation under incomplete product availability. *Amer. Econom. J. Microeconom.* 5(4): 1–30.

Ferreira KJ, Parthasarathy S, Sekar S (2020) Learning to rank an assortment of products. Preprint, submitted June 15, https://dx.doi.org/10.2139/ssrn.3395992.

Fisher ML, Hammond JH, Obermeyer WR, Raman A (1994) Making supply meet demand in an uncertain world. *Harvard Bus. Rev.* 72(3):83–93.

Fisher M, Vaidyanathan R (2014) A demand estimation procedure for retail assortment optimization with results from implementations. *Management Sci.* 60(10):2401–2415.

Fitzsimons GJ (2000) Consumer response to stockouts. *J. Consumer Res.* 27(2):249–266.

Gallino S, Moreno A (2014) Integration of online and offline channels in retail: The impact of sharing reliable inventory availability information. *Management Sci.* 60(6):1434–1451.

Gruen T, Corsten D (2008) A comprehensive guide to retail out-of-stock reduction in the fast-moving consumer goods industry. Proctor & Gamble Company research study.

Gruen T, Corsten D, Bharadwaj S (2002) *Retail Out-of-Stocks: A Worldwide Examination of Causes, Rates, and Consumer Responses* (Grocery Manufacturers of America, Washington, DC).

Huang Y, Zhang YC (2016) The out-of-stock (OOS) effect on choice shares of available options. *J. Retailing* 92(1):13–24.

Jiang Z, Li J, Zhang D (2020) A high-dimensional choice model for online retailing. Preprint, submitted September 17, https://dx.doi.org/10.2139/ssrn.3687727.

Jing X, Lewis M (2011) Stockouts in online retailing. *J. Marketing Res.* 48(2):342–354.

Kalyanam K, Borle S, Boatwright P (2007) Deconstructing each item's category contribution. *Marketing Sci.* 26(3):327–341.

Karmarkar US, Patel NR (1977) The one-period, N-location distribution problem. *Naval Res. Logist. Quart.* 24(4):559–575.

Li S, Chen R, Yang L, Huang D, Huang S (2020) Predictive modeling of consumer color preference: Using retail data and merchandise images. *J. Forecasting* 39(8):1305–1323.

Lissaman C (2019) The size of the global fashion market. Accessed February 14, 2022, https://www.commonobjective.co/article/the-size-of-the-global-fashion-retail-market.

Musalem A, Olivares M, Bradlow ET, Terwiesch C, Corsten D (2010) Structural estimation of the effect of out-of-stocks. *Management Sci.* 56(7):1180–1197.

Paterson C, Kiesmüller G, Teunter R, Glazebrook K (2011) Inventory models with lateral transshipments: A review. *Eur. J. Oper. Res.* 210(2):125–136.

Sanchez-Ruiz L, Blanco B, Kyguolienė A (2018) A theoretical overview of the stockout problem in retail: From causes to consequences. *Management Organ. Systematic Res.* 79(1):103–116.

Smith SA, Achabal DD (1998) Clearance pricing and inventory policies for retail chains. *Management Sci.* 44(3):285–300.

Vulcano G, Van Ryzin G, Ratliff R (2012) Estimating primary demand for substitutable products from sales transaction data. *Oper. Res.* 60(2):313–334.